Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/15/2020
PI Name:	Menon, Jayant M.D., M.Eng.		
Project Title:	Autonomous Medical Response Agent (AMRA)		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Human Research Program Elements:	None		
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Space Biology Element:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2019 (originally 8/31/2019) per E. Urquieta/TRISH (Ed., 9/18/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Menon, P.K. Ph.D. (Optimal Synthesis)		
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Performance Goal Text:			
Task Description:	Industry Project Autonomous Medical Response Agent (AMRA) is an autonomous medical response agent that assists patients in self-diagnosis and treatment while coordinating care with remote specialists. The response agent derives a patient's present state vector with a branching medical history question tree that incorporates qualitative and quantitative data. This anomaly vector prompts a prioritized set of control actions from published medical literature, and the results of these control actions update the present state vector until the desired healthy state is achieved. AMRA's recorded timeline of control actions and results uses internationally recognized codes of disease and treatment; it is explainable, accountable, and Electronic Medical Record (EMR) interoperable.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Key Findings: 1) A distinct Physician/Remote Flight Surgeon GUI (graphical user interface) was developed after human-centered research determined it necessary to display Bayesian computed ranking to Expert users. This builds trust and confidence in the behavior of AMRA and enables rapid communication to onboard and remote experts for coordinated medical care. (HARI-02: We need to develop design guidelines for effective human-automation-robotic systems in operational environments that may include distributed, non-colocated adaptive mixed-agent teams with variable transmission latencies.). 2) Multiple Astronaut and Flight Surgeon users desired personalized medical recommendations in an autonomous agent. A weighted utility calculation was developed based on the individual preference for strength of medical evidence, current mission inventory, invasiveness of treatment, time to treatment effect, and cost of treatment (e.g., side effect). This enables AMRA to provide evidence-based personalized care for astronauts. (Med-03: We do not know how to apply personalized medicine effectively to reduce health risk for a selected crew.). 3) A prototype database implementation was employed to provide a single data architecture in which to store past and present inputs from the astronaut as well as vehicle health data. This personal medical record includes all state descriptions of the patient to provide a rigorous account of all medical decisions made. This will enable AMRA to integrate disparate tools into a seamless experience; one that merges inventory, scheduling, just in time expertise, on-board medical devices, and environmental monitoring. 4) AMRA demonstrates the potential of integrating supervised (Bayesian, labeled data sets) with un-supervised (Speech recognition engine) machine learning techniques. Impact: The presented work demonstrates it is possible to design a software architecture and algorithm that can provide explainable, relevant, adaptive, integrated and team-based autonomous medical decision support for Astronauts on long duration missions. The AMRA prototype provides a path to develop safe autonomous medical decision support.
Task Progress:	Autonomous Medical Response Agent (AMRA) is an autonomous computed medical decision support tool that assists astronauts on deep space missions with diagnosis and personalized treatment while coordinating care with remote distributed medical experts with variable transmission latencies. AMRA addresses Human Research Program (HRP) Gaps Med-10, Med-03, Med-09, and HARI-02 (Med10: We do not have the capability to provide computed medical decision support during exploration missions; Med03: We do not know how to apply personalized medicine effectively to reduce health risk for a selected crew; Med09: We do not have the capability to predict estimated medical risk posture during exploration missions based on current crew health and resources; HARI-02: We need to develop design guidelines for effective human-automation-robotic systems in operational environments that may include distributed, non-colocated adaptive mixed-agent teams with variable transmission latencies). Through an Astronaut user interface, AMRA guides a medical history, physical exam and diagnostic tests, incorporating qualitative and quantitative data, to compute the rank of various medical conditions by likelihood and predicted level of disability (Med-10). This allows AMRA to prioritize the management of the the medical condition that is most likely and most disabling. An organ system based control algorithm guides specific therapeutic actions defined by NASA Space operations literature (ISS Medical Manual) on likely condition and predicted outcome, and personal preferences (Med-03). Feedback from observed treatment outcomes is compared with desired outcomes (NASA STD 3001 v1) and provides an estimated risk to crew and mission (Med-09). The resulting difference between observed outcome and desired outcome drives continuous decision support, allowing AMRA to provide personalized continuity of care that bridges time delays and intermittent connectivity with Earth based mission support. When a connection is established, AMRA autonomously transmits individual crew member health states to remote flight surgeons providing asynchronous coordination with non-colocated mission support teams (HARI-02). AMRA's multi criteria Bayesian feedback control system records all rankings, recommended actions, and recorded treatment outcomes in an explainable, accountable, and interoperable electronic health database as a part of NASA's Medical Decision Architecture. The prototype is demonstrated as being integrated with data from both onboard commercial medical diagnostic equipment and environmental monitoring databases (e.g., ECLSS--Environmental Control and Life Support System). AMRA can be used for Astronaut self-care, respecting the stated wish for patient privacy on long missions, as well as to guide Astronauts to care for other crew members (e.g., Cardiopulmonary Resuscitation). Ultimately, AMRA can provide long term, continuous, autonomous medical guidance for the entire crew. Completed Project Aims: 1) Human Centered Design research to prioritize AMRA prototype for two high-impact diseases from the HRP Medical Conditions List: Headache and Difficulty Breathing ; 2) Define architecture of medical decision making including triage, medical history, physical exam, diagnostic tests, and treatment recommendations for various Space based causes for Headache and Difficulty Breathing; 3) Define elements of medical outcomes based feedback control for the medical management of Headache and Difficulty breathing; 4) AMRA GUI for Astronauts with software infrastructure that functions fully onboard without Earth connectivity. This aim was completed with assistance from NASA Flight Surgeons and Astronauts; 5) Further develop multicriteria Bayesian computed decision support algorithm for autonomous medical decision support. (Med-10); 6) Prototype speech interface for the Astronaut GUI increases usability when users need both hands to complete diagnostic and therapeutic tasks. Key Findings: 1) A distinct Physician/Remote Flight Surgeon GUI was developed after human-centered research determined it necessary to display Bayesian computed ranking to Expert users. This builds trust and confidence in the behavior of AMRA and enables rapid communication to onboard and remote experts for coordinated medical care. (HARI-02). 2) Multiple Astronaut and Flight Surgeon users desired personalized medical recommendations in an autonomous agent. A weighted utility calculation was developed based on the individual preference for strength of medical evidence, current mission inventory, invasiveness of treatment, time to treatment effect, and cost of treatment (e.g., side effect). This enables AMRA to provide evidence-based personalized care for astronauts. (Med-03). 3) A full database implementation was necessary to provide coordinate an electronic medical record of autonomous medical decision support with NASA's Medical Data Architecture. This enables AMRA to integrate with tools that manage inventory, scheduling, and logistics, on-board medical devices, and environmental monitoring. 4) AMRA demonstrates the potential of integrating supervised (Bayesian) with un-supervised (Speech engine) machine learning techniques. Impact: The presented work demonstrates it is possible to design a software architecture and algorithm that can provide explainable, relevant, adaptive, integrated, and team-based autonomous medical decision support for Astronauts on long duration missions. The AMRA prototype provides a path to develop safe autonomous medical decision support. Proposed Future Research: 1) Situate decisions in an overall mission timeline representative of operational constraints of asynchronous coordination and continuity of control with Earth-based mission support center and autonomy when connectivity is unavailable. 2) Expand the decision support capabilities of AMRA to include guidance for medical and

	non-medical decision makers with successful demonstration in an analogous environment that is relevant to NASA's exploration missions. 3) Deeper integration of Bayesian decision systems with traditional machine learning tools to extend the scale and speed of implementation of AMRA.
Bibliography Type:	Description: (Last Updated:)